

Last Change	Review	Approval
Joern Huber Digital unterschrieben von Joern Huber Datum: 2026.08.31 12:23:10 +02'00'	Klaus Eickel Digital unterschrieben von Klaus Eickel Datum: 2026.09.01 13:46:46 +02'00'	Stephan Zidowitz Digitally signed by Stephan Zidowitz Date: 2026.09.01 15:00:14 +02'00'

Document History

Version	Date	Changes	JIRA-Ticket	Name
V1	2026-08-31	Release of v1.0 of TACTIX MRI Quality Assessment	n/a	jhu

 Fraunhofer MEVIS	TACTIX MRI QUALITY ASSESSMENT	Document-ID TACTIX_MRIQA	Last Change by jhu
	Release Letter	V1	Last Change on 2026-08-31
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1 Introduction and Overview

This document summarizes the delivery of the **TACTIX MRI Quality Assessment software**. The software delivers a Streamlit-based tutorial application for MRI image quality assessment, enabling users to upload MR/DICOM images and interactively evaluate key quality metrics including Universal Quality Index (UQI), sharpness, and signal-to-noise ratio (SNR). It combines practical metric computation with transparent, step-by-step explanations of core image-processing methods to support reproducible quality evaluation and user training.

Contact person at Fraunhofer MEVIS: Klaus Eickel (klaus.eickel@mevis.fraunhofer.de)

2 Release Summary

Software Component	TACTIX MRI Quality Assessment
Version	https://gitlab.fme.lan/tactix/tactix-tutorials/-/tags/tactix_mri_quality_tutorial_v1.0 SHA: 682cc5b0ccd5dc86cdc0ed1fd6ca91baa0323ee3
Embedding Product(s)	Not applicable
Purpose of this Release	Provision of the TACTIX MRI Quality Assessment software under MIT license for research and educational purposes. Not for clinical use.
OS and Compiler Details	OS: Windows 11 Python Version: 3.12
Embedded Libraries	The following libraries are used by the software. Detailed license information is contained in the delivered "third_party_licenses" folder and in the NOTICE file. - streamlit - Numpy - Scikit-image - Matplotlib - Pydicom - imageio
Functionality	The software provides a Streamlit-based educational application for Magnetic Resonance Image (MRI) quality assessment. It provides an interactive, guided workflow to upload MR/DICOM images, compute key quality metrics (Universal Quality Index, sharpness, and signal-to-noise ratio), and understand how these values are derived through visual and step-by-step explanations. The project combines practical metric calculation with foundational image-processing concepts (e.g., thresholding, filtering, morphology, and entropy) to support transparent, reproducible quality evaluation and training.
Regulatory Compliance	THIS SOFTWARE IS PROVIDED "AS IS" AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED.

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	In particular, Fraunhofer MEVIS does not ensure compliance to medical product regulations for the release.
Safety Classification	None
Risk Measures	No risk measures defined.
Known Nonconformities and Restrictions	The software is not qualified for use as a medical product or as part thereof. No bugs or restrictions are known. This software was (and is continuously) developed in the context of an EU-funded project and thus this release reflects the current state of research without having undergone comprehensive validation. We therefore do not guarantee correctness and conformities of any included algorithms. If you notice any incorrectness, bugs or faulty results, you may report those to the project maintainers (contact mail see above).